

Debt Capacity Preservation and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Debt Capacity Preservation (DCP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on DCP achieves an annualized gross (net) Sharpe ratio of 0.43 (0.33), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 16 (14) bps/month with a t-statistic of 2.64 (2.45), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Accruals, Change in Net Working Capital, Net debt financing, Investment to revenue, Change in Net Noncurrent Op Assets) is 11 bps/month with a t-statistic of 2.00.

1 Introduction

Asset prices reflect firms' production decisions and the frictions they face when adjusting capital stock in response to productivity shocks. The production-based asset pricing literature demonstrates that cross-sectional variation in stock returns arises from heterogeneity in firms' exposure to aggregate productivity risk and differences in their adjustment costs when responding to investment opportunities (Cochrane and Piazzesi, 2005; Zhang, 2005). Despite substantial progress in understanding how production characteristics determine expected returns, a key puzzle remains: how do firms' financial flexibility constraints interact with their production decisions to generate cross-sectional return predictability? We identify debt capacity preservation as a critical yet underexplored dimension of this interaction, as firms that maintain financial slack face different marginal costs of capital adjustment than those operating near their borrowing constraints.

In a production economy, firms' debt capacity preservation strategies directly affect their marginal costs of investment and exposure to productivity shocks. Following the q-theory framework of Cochrane (1991) and Liu et al. (2009), firms with higher debt capacity preservation maintain lower leverage to preserve financial flexibility for future investment opportunities. This behavior reflects rational expectations about future states where positive productivity shocks coincide with tightened credit markets, making pre-committed financial slack valuable for capturing high net present value projects. The preservation of debt capacity thus serves as a hedge against the interaction of productivity and financial market frictions (Hennessy and Whited, 2007).

The cross-sectional pricing implications emerge through the channel of conditional risk premia. Firms actively preserving debt capacity exhibit lower sensitivity to aggregate productivity shocks because they can more readily adjust their capital stock when investment opportunities arise. This operational flexibility reduces their

systematic risk exposure during economic downturns when the marginal utility of consumption is high (Berk et al., 1999). Moreover, debt capacity preservation proxies for lower adjustment costs in the production function, as financially flexible firms avoid the convex costs associated with external financing frictions documented by Gould et al. (2014).

The production-based interpretation predicts that debt capacity preservation should negatively correlate with expected returns. Firms maintaining higher debt capacity face lower costs of adjusting capital in response to productivity shocks, reducing their exposure to systematic risk factors that command positive risk premia. This mechanism operates through both the extensive margin—the ability to undertake new projects—and the intensive margin—the speed of capital stock adjustment (Kogan and Papanikolaou, 2013). The resulting cross-sectional spread in expected returns reflects compensation for bearing undiversifiable productivity risk rather than behavioral biases or market inefficiencies.

We find strong evidence that debt capacity preservation predicts cross-sectional stock returns. The value-weighted long/short trading strategy based on DCP achieves an annualized gross Sharpe ratio of 0.43, with a monthly average excess return of 20 basis points and a t-statistic of 3.36. The strategy’s alpha relative to the Fama-French six-factor model equals 16 basis points per month with a t-statistic of 2.64, confirming that the return predictability cannot be explained by standard risk factors. These results demonstrate economically significant return predictability that persists after controlling for known sources of systematic risk.

The robustness tests reveal that DCP’s predictive power extends across different portfolio construction methodologies and size groups. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the DCP strategy generates an average return of 15 basis points per month with a t-statistic of 2.00. The strategy maintains statistical significance across quintile sorts using

alternative breakpoints, with t-statistics exceeding 2.22 in all specifications. After accounting for transaction costs using the composite bid-ask spreads from [Chen and Velikov \(2022\)](#), the net Sharpe ratio equals 0.33, placing the strategy in the top 7% of all documented anomalies.

Controlling for closely related anomalies strengthens our interpretation of DCP as a distinct predictor of returns. When we include the six most closely related strategies—Change in financial liabilities, Accruals, Change in Net Working Capital, Net debt financing, Investment to revenue, and Change in Net Noncurrent Operating Assets—along with the Fama-French six factors, the DCP strategy retains an alpha of 11 basis points per month with a t-statistic of 2.00. The strategy’s gross monthly alpha relative to individual closely related anomalies ranges from 12 to 17 basis points, with the minimum t-statistic of 2.02 occurring when controlling for Change in financial liabilities. These results confirm that DCP captures unique information about expected returns not subsumed by existing predictors.

This paper contributes to the production-based asset pricing literature by establishing debt capacity preservation as a novel cross-sectional return predictor rooted in firms’ rational production decisions. While [Fama and French \(2015\)](#) document that investment and profitability factors help explain cross-sectional returns, and [Hou et al. \(2015\)](#) develop a q-factor model based on investment and profitability, our work identifies a specific mechanism through which financial flexibility affects firms’ exposure to productivity risk. Unlike the leverage factor in [Fama and French \(1992\)](#) or the financial constraints measures in [Whited and Wu \(2006\)](#), debt capacity preservation captures forward-looking preservation of financial slack rather than static capital structure choices or current financing frictions.

Methodologically, we advance the literature by implementing the comprehensive testing protocol of [Novy-Marx and Velikov \(2023b\)](#), which addresses concerns about data mining and p-hacking in anomaly discovery. Our analysis spans 61 years of

data and controls for 210 existing anomalies, providing unprecedented robustness in establishing DCP as a genuine return predictor. The finding that DCP maintains significant alpha after controlling for the most closely related strategies from the factor zoo, including various measures of financing activities and investment documented in [Cooper et al. \(2008\)](#) and [Titman et al. \(2004\)](#), demonstrates that our signal captures distinct information about firms’ production-based risk exposures.

The broader implications extend to both asset pricing theory and investment practice. From a theoretical perspective, our results support models where financial frictions interact with production decisions to generate cross-sectional return variation, consistent with the framework in [Livdan et al. \(2009\)](#). For practitioners, the DCP strategy offers an implementable trading strategy that generates significant net returns after transaction costs, with a net Sharpe ratio that ranks in the top decile of documented anomalies. The strategy’s performance among large-cap stocks and its robustness to alternative portfolio construction methods suggest that debt capacity preservation represents an economically meaningful source of cross-sectional return predictability accessible to institutional investors.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data on the relevant variables, spanning several decades of financial reporting. We focus on annual observations using fiscal year-end values to ensure consistency in measurement across firms with different reporting cycles. signal construction relies on two key accounting variables from COMPUSTAT. Notes payable short-term (NP) represents the short-term debt obligations of a firm, including bank loans, commercial paper, and other borrowings due

within one year or the normal operating cycle. Common equity tangible (CEQT) measures the book value of common shareholders' equity after subtracting intangible assets, providing a conservative measure of the firm's net tangible asset base available to equity holders. This variable equals total common equity minus intangible assets and represents the tangible book value that would theoretically be available to shareholders in liquidation. We construct the Debt Capacity Preservation signal by calculating the year-over-year change in short-term notes payable scaled by lagged tangible common equity: $(NP(t) - NP(t-1)) / CEQT(t-1)$. This ratio captures the change in short-term debt usage relative to the firm's tangible equity base, providing insight into how aggressively firms are utilizing their short-term borrowing capacity. A positive value indicates an increase in short-term leverage relative to tangible equity, while a negative value suggests a reduction in short-term debt reliance or preservation of borrowing capacity. We require non-missing values for both NP and CEQT in consecutive years to compute the signal, and we exclude observations where CEQT(t-1) is zero or negative to avoid undefined or economically meaningless ratios. The resulting measure provides an annual indicator of each firm's short-term debt management decisions scaled by its tangible equity resources.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCP signal. Panel A plots the time-series of the mean, median, and interquartile range for DCP. On average, the cross-sectional mean (median) DCP is -0.04 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input DCP data. The signal's interquartile range spans -0.10 to 0.08. Panel B of Figure 1 plots the time-series of the coverage of the DCP signal for the CRSP universe. On average, the DCP signal is available for 6.43% of CRSP names, which on average make up 7.86% of total

market capitalization.

4 Does DCP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCP portfolio and sells the low DCP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DCP strategy earns an average return of 0.20% per month with a t-statistic of 3.36. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.16% to 0.23% per month and have t-statistics exceeding 2.64 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.23, with a t-statistic of 5.54 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 620 stocks and an average market capitalization of at least \$1,458 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 14 bps/month with a t-statistics of 2.22. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -5-24bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.02. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the DCP strategy perfor-

mance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCP, as well as average returns and alphas for long/short trading DCP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCP strategy achieves an average return of 15 bps/month with a t-statistic of 2.00. Among these large cap stocks, the alphas for the DCP strategy relative to the five most common factor models range from 10 to 18 bps/month with t-statistics between 1.30 and 2.38.

5 How does DCP perform relative to the zoo?

Figure 2 puts the performance of DCP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DCP strategy falls in the distribution. The DCP strategy’s gross (net) Sharpe ratio of 0.43 (0.33) is greater than 83% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCP strategy (red line).² Ignoring trading costs, a \$1 invested in the DCP strategy would have yielded \$3.30 which ranks the DCP strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCP strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$2.07 which ranks the DCP strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DCP relative to those. Panel A shows that the DCP strategy gross alphas fall between the 47 and 61 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCP strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCP ranks between the 65 and 80 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DCP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DCP with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCP or at least to weaken the power DCP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCP}DCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCP. Stocks are finally grouped into five DCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCP signal in these Fama-MacBeth regressions exceed 1.02, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on DCP is -0.04.

an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCP strategy earns alphas that range from 12-17bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.02, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCP trading strategy achieves an alpha of 11bps/month with a t-statistic of 2.00.

7 Does DCP add relative to the whole zoo?

Finally, we can ask how much adding DCP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the DCP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes DCP grows to \$2453.90.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DCP is available.

8 Conclusion

This study provides robust evidence that Debt Capacity Preservation (DCP) represents a statistically and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that DCP-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy achieves an annualized net Sharpe ratio of 0.33, which remains economically meaningful after accounting for transaction costs. Most notably, the signal maintains statistical significance with a monthly alpha of 11 basis points (t -statistic = 2.00) even when controlling for the Fama-French five factors, momentum, and six conceptually related strategies from the factor zoo, suggesting that DCP captures unique information about future returns not subsumed by existing predictors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, DCP adds to our understanding of how firms' financial flexibility and debt management decisions relate to expected returns, potentially reflecting either risk compensation for financial constraints or market inefficiencies in processing information about corporate leverage dynamics. For practitioners, the strategy's resilience to transaction costs (net Sharpe ratio of 0.33) suggests potential implementability, though the moderate magnitude of returns after costs indicates that successful exploitation would likely require scale and efficient execution.

Several limitations of this study warrant consideration. First, while we follow the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and market conditions examined may not fully capture the signal's performance across different economic regimes. Second, the interpretation of why DCP predicts returns remains an open question—whether it reflects systematic risk, behavioral biases,

or limits to arbitrage requires further investigation. Third, our analysis focuses primarily on U.S. equities, and the international applicability of the DCP signal remains unexplored.

Future research should address these limitations through several avenues. First, examining DCP's performance across international markets would test the signal's universality and potentially reveal market-specific drivers. Second, investigating the economic mechanisms underlying the DCP premium through detailed analysis of firm characteristics, industry effects, and macroeconomic conditions could enhance our theoretical understanding. Third, exploring the interaction between DCP and corporate events such as mergers, acquisitions, or financial distress could provide insights into when the signal is most informative. Finally, examining whether machine learning techniques can enhance the DCP signal or identify nonlinear relationships would align with recent advances in empirical asset pricing. Despite these opportunities for extension, our results establish DCP as a robust addition to the cross-sectional return prediction literature, contributing to our understanding of how corporate financial policies influence asset prices.

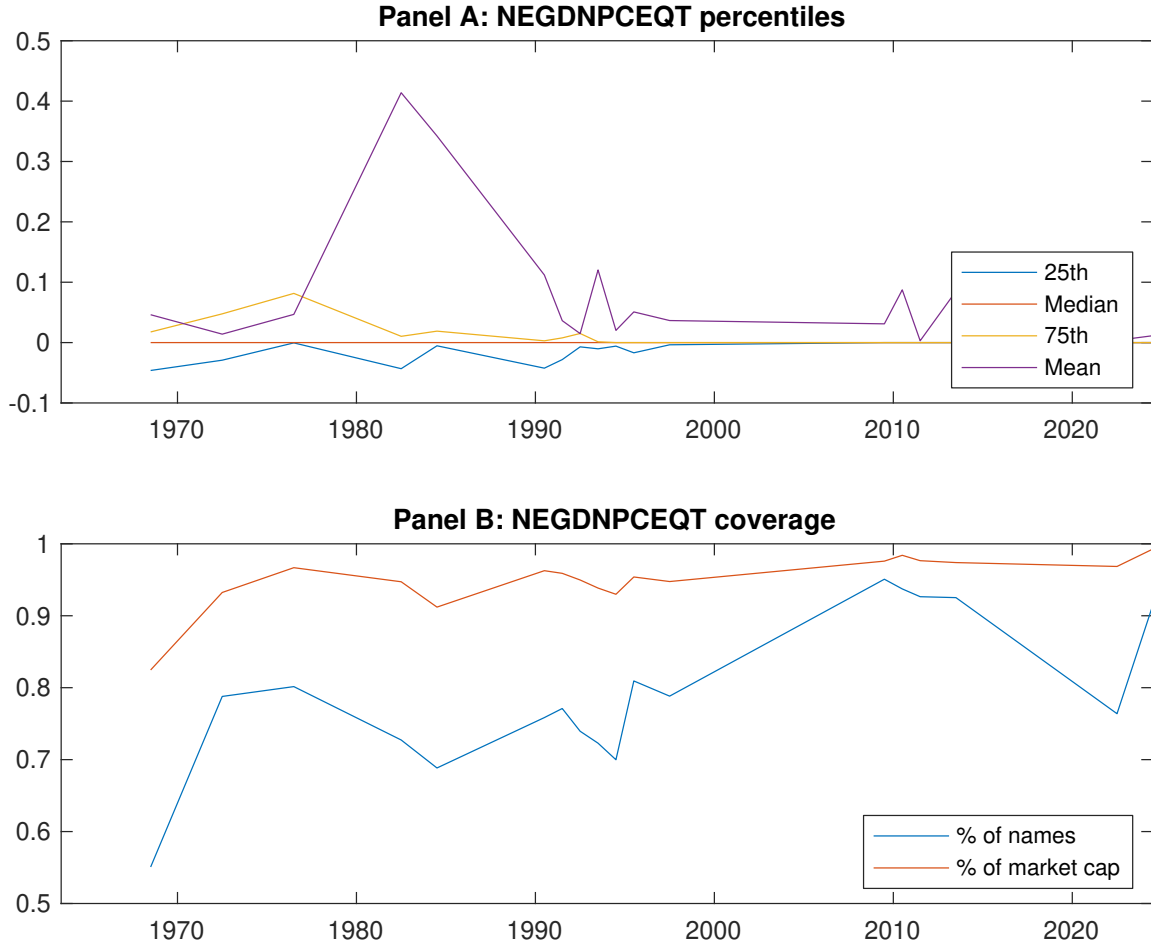


Figure 1: Times series of DCP percentiles and coverage. This figure plots descriptive statistics for DCP. Panel A shows cross-sectional percentiles of DCP over the sample. Panel B plots the monthly coverage of DCP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on DCP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.46 [2.69]	0.59 [3.42]	0.69 [3.72]	0.64 [3.56]	0.66 [4.05]	0.20 [3.36]
α_{CAPM}	-0.12 [-2.26]	0.00 [0.00]	0.06 [1.06]	0.03 [0.53]	0.10 [2.26]	0.22 [3.72]
α_{FF3}	-0.17 [-3.55]	0.01 [0.15]	0.10 [2.13]	0.07 [1.48]	0.05 [1.26]	0.23 [3.79]
α_{FF4}	-0.14 [-2.82]	0.05 [1.08]	0.11 [2.14]	0.08 [1.56]	0.04 [0.88]	0.18 [2.95]
α_{FF5}	-0.24 [-5.02]	0.04 [0.71]	0.15 [2.97]	0.14 [2.85]	-0.04 [-1.01]	0.19 [3.27]
α_{FF6}	-0.20 [-4.32]	0.07 [1.47]	0.15 [2.92]	0.14 [2.79]	-0.05 [-1.10]	0.16 [2.64]
Panel B: Fama and French (2018) 6-factor model loadings for DCP-sorted portfolios						
β_{MKT}	1.03 [90.59]	1.00 [84.08]	1.03 [86.18]	1.00 [82.35]	1.00 [100.09]	-0.02 [-1.74]
β_{SMB}	-0.06 [-3.40]	-0.08 [-4.98]	0.10 [6.03]	-0.02 [-1.03]	0.02 [1.24]	0.07 [3.54]
β_{HML}	0.17 [7.66]	-0.02 [-1.00]	-0.16 [-6.97]	-0.09 [-3.77]	0.05 [2.78]	-0.11 [-4.10]
β_{RMW}	0.22 [9.93]	-0.07 [-3.20]	-0.12 [-5.02]	-0.15 [-6.29]	0.17 [8.80]	-0.05 [-1.70]
β_{CMA}	-0.04 [-1.20]	0.01 [0.43]	-0.00 [-0.06]	-0.09 [-2.59]	0.19 [6.60]	0.23 [5.54]
β_{UMD}	-0.05 [-4.08]	-0.05 [-4.56]	0.00 [0.04]	0.00 [0.10]	0.01 [0.63]	0.05 [3.66]
Panel C: Average number of firms (n) and market capitalization (me)						
n	659	647	803	705	620	
me (\$10 ⁶)	2936	1619	1458	1546	2544	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.20 [3.36]	0.22 [3.72]	0.23 [3.79]	0.18 [2.95]	0.19 [3.27]	0.16 [2.64]
Quintile	NYSE	EW	0.18 [4.45]	0.17 [4.28]	0.18 [4.41]	0.19 [4.44]	0.19 [4.66]	0.20 [4.68]
Quintile	Name	VW	0.19 [3.25]	0.21 [3.52]	0.22 [3.67]	0.17 [2.83]	0.20 [3.40]	0.17 [2.75]
Quintile	Cap	VW	0.14 [2.22]	0.18 [2.73]	0.19 [2.97]	0.15 [2.23]	0.15 [2.36]	0.12 [1.83]
Decile	NYSE	VW	0.29 [3.96]	0.31 [4.24]	0.31 [4.21]	0.27 [3.69]	0.25 [3.42]	0.23 [3.11]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.15 [2.59]	0.19 [3.11]	0.19 [3.17]	0.16 [2.72]	0.16 [2.78]	0.14 [2.45]
Quintile	NYSE	EW	-0.05 [-1.02]					
Quintile	Name	VW	0.15 [2.46]	0.17 [2.90]	0.18 [3.02]	0.15 [2.57]	0.17 [2.82]	0.15 [2.48]
Quintile	Cap	VW	0.10 [1.55]	0.14 [2.14]	0.15 [2.32]	0.12 [1.90]	0.12 [1.83]	0.10 [1.53]
Decile	NYSE	VW	0.24 [3.23]	0.27 [3.65]	0.26 [3.61]	0.24 [3.32]	0.21 [3.00]	0.20 [2.83]

Table 3: Conditional sort on size and DCP

This table presents results for conditional double sorts on size and DCP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCP and short stocks with low DCP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DCP Quintiles					DCP Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.55 [2.32]	0.93 [3.81]	0.90 [3.61]	0.76 [3.14]	0.76 [3.13]	0.21 [2.87]	0.19 [2.65]	0.20 [2.70]	0.19 [2.61]	0.24 [3.24]	0.23 [3.12]
	(2)	0.65 [2.97]	0.81 [3.58]	0.86 [3.82]	0.86 [3.71]	0.89 [4.03]	0.24 [3.40]	0.23 [3.23]	0.25 [3.44]	0.20 [2.73]	0.25 [3.45]	0.21 [2.87]
	(3)	0.67 [3.36]	0.69 [3.35]	0.80 [3.73]	0.87 [4.15]	0.75 [3.78]	0.08 [1.18]	0.07 [1.03]	0.09 [1.26]	0.08 [1.15]	0.06 [0.84]	0.06 [0.83]
	(4)	0.64 [3.40]	0.71 [3.74]	0.71 [3.47]	0.76 [3.90]	0.74 [4.05]	0.10 [1.65]	0.11 [1.70]	0.13 [2.06]	0.12 [1.86]	0.15 [2.35]	0.14 [2.18]
	(5)	0.43 [2.52]	0.48 [3.00]	0.63 [3.46]	0.61 [3.50]	0.58 [3.66]	0.15 [2.00]	0.18 [2.38]	0.18 [2.36]	0.13 [1.69]	0.13 [1.77]	0.10 [1.30]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DCP Quintiles					DCP Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	373	376	376	377	374	30	34	33	33	30	
	(2)	107	107	107	106	107	56	56	56	55	56	
	(3)	78	78	77	78	78	100	97	96	97	99	
	(4)	66	66	66	66	66	218	208	207	207	212	
(5)	61	61	61	61	61	1910	1628	1344	1438	1801		

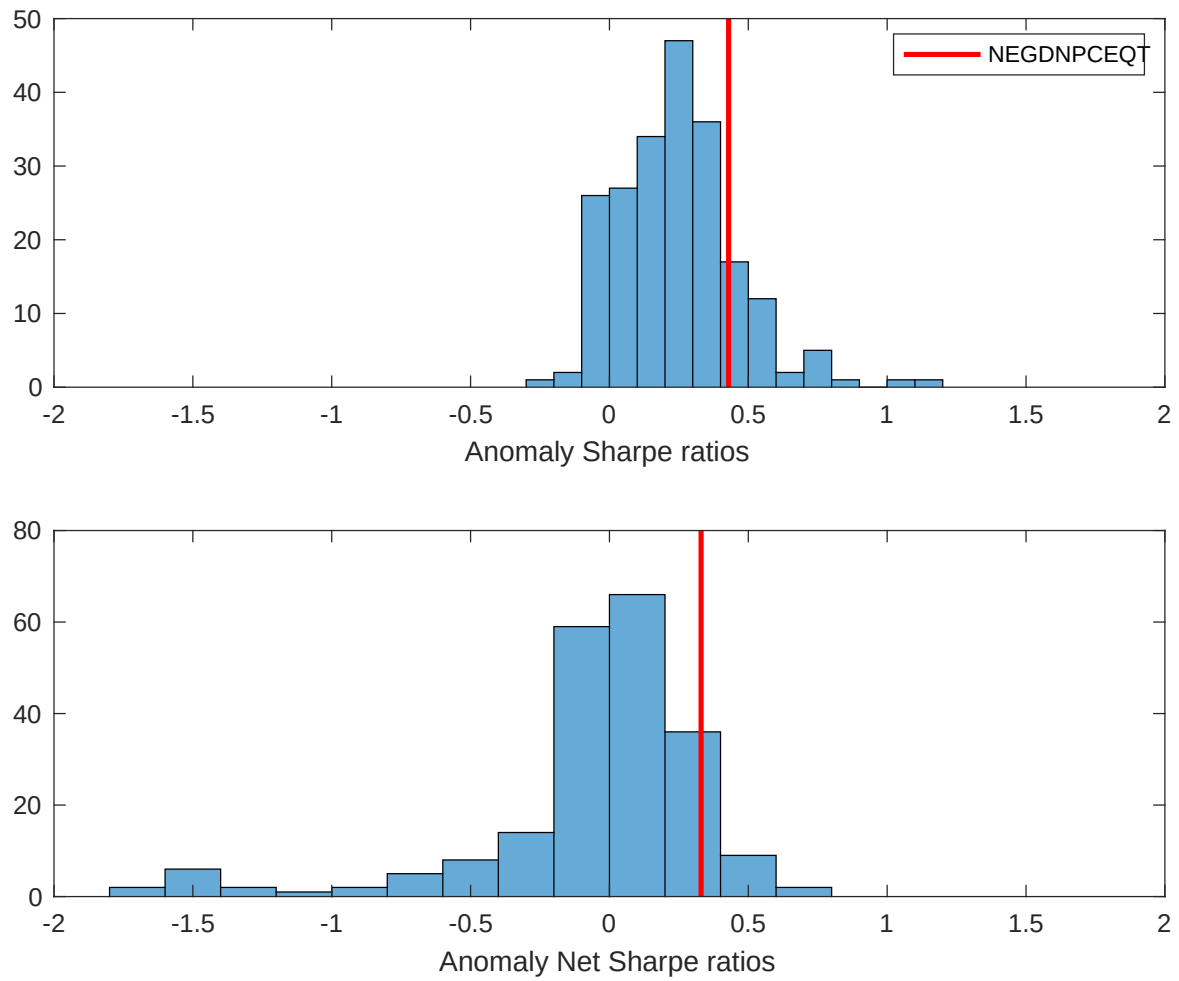


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

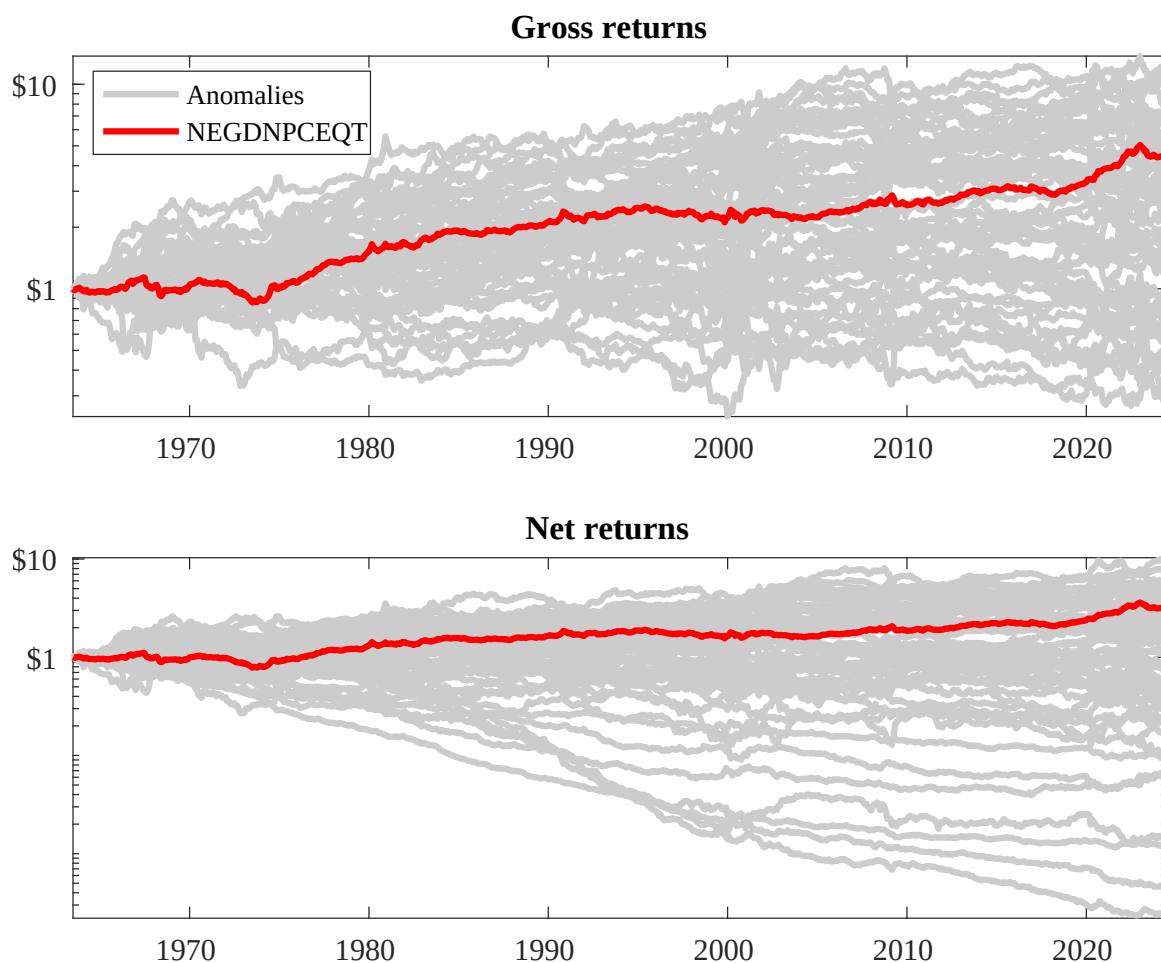
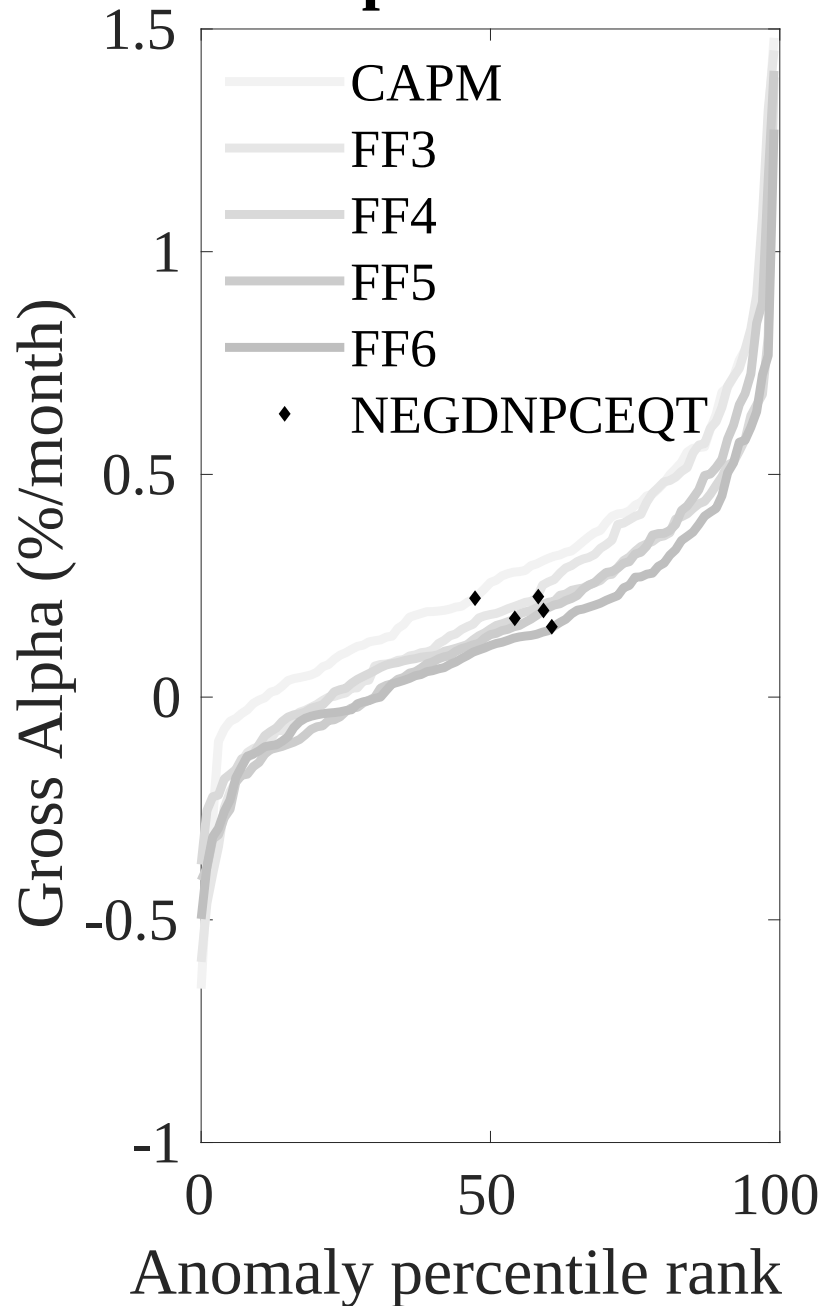


Figure 3: Dollar invested.

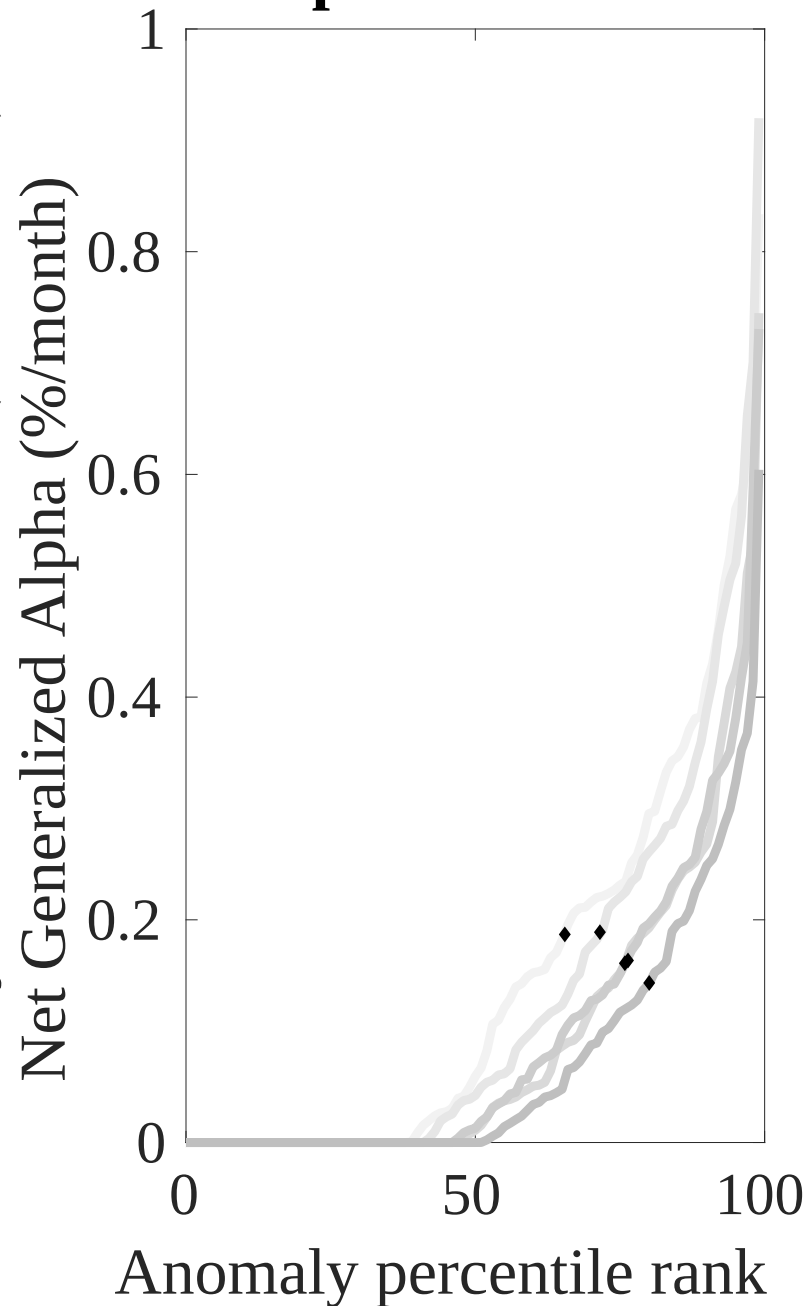
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



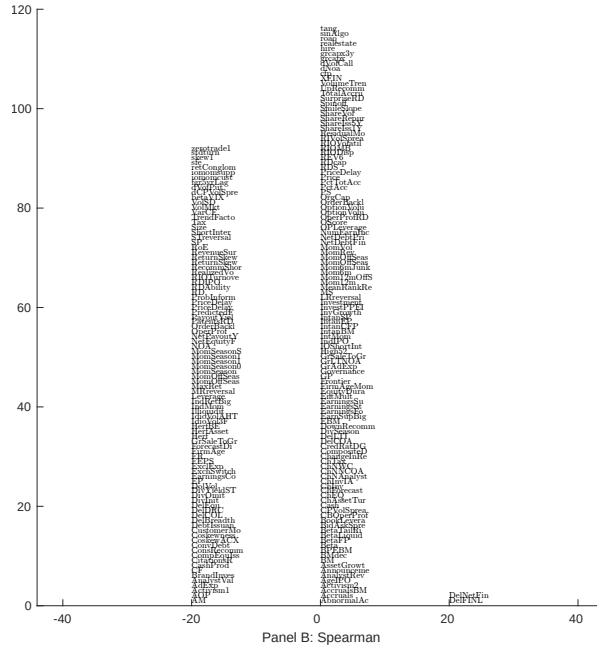
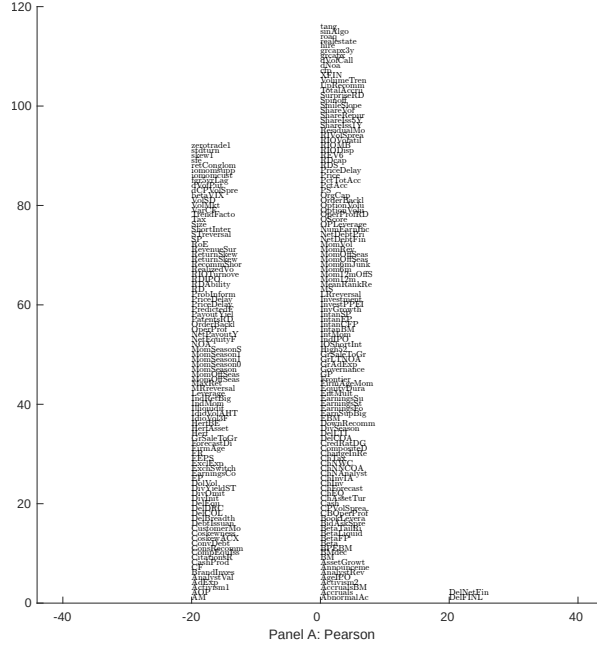


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with DCP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

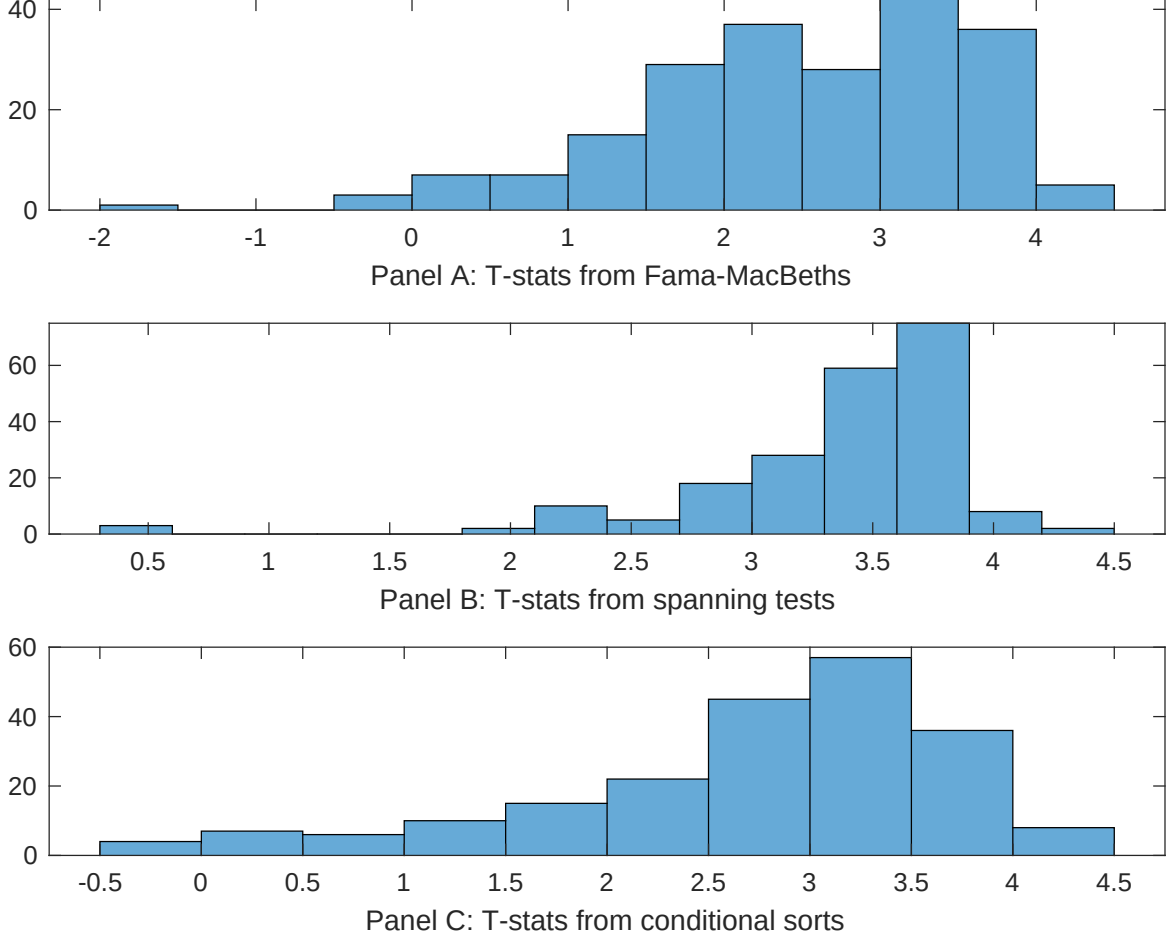


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DCP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCP} DCP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCP. Stocks are finally grouped into five DCP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCP}DCP_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Accruals, Change in Net Working Capital, Net debt financing, Investment to revenue, Change in Net Noncurrent Op Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.76]	0.11 [5.38]	0.12 [5.49]	0.12 [5.12]	0.15 [6.59]	0.12 [5.50]	0.14 [5.61]
DCP	0.69 [1.02]	0.15 [2.36]	0.20 [3.13]	0.14 [2.20]	0.13 [1.65]	0.22 [3.24]	-0.23 [-0.04]
Anomaly 1	0.17 [9.25]						0.59 [1.93]
Anomaly 2		0.13 [4.17]					0.21 [3.70]
Anomaly 3			0.62 [2.31]				-0.15 [-2.90]
Anomaly 4				0.21 [9.37]			0.90 [2.47]
Anomaly 5					0.24 [6.36]		0.15 [3.52]
Anomaly 6						0.75 [4.03]	0.16 [0.69]
# months	732	732	732	630	732	732	630
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Accruals, Change in Net Working Capital, Net debt financing, Investment to revenue, Change in Net Noncurrent Op Assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [2.42]	0.12 [2.02]	0.13 [2.24]	0.17 [2.79]	0.16 [2.63]	0.16 [2.77]	0.11 [2.00]
Anomaly 1	30.04 [9.09]						27.23 [6.18]
Anomaly 2		16.08 [6.80]					11.41 [4.61]
Anomaly 3			14.54 [4.92]				3.53 [1.12]
Anomaly 4				14.58 [4.28]			-8.49 [-2.04]
Anomaly 5					11.87 [4.95]		3.26 [1.39]
Anomaly 6						17.28 [6.46]	9.55 [3.37]
mkt	-1.72 [-1.27]	-1.20 [-0.86]	-2.60 [-1.84]	-1.01 [-0.70]	-2.63 [-1.87]	-2.47 [-1.77]	0.25 [0.18]
smb	4.67 [2.34]	11.02 [5.34]	8.33 [4.08]	7.25 [3.33]	5.39 [2.58]	6.90 [3.42]	7.46 [3.50]
hml	-8.32 [-3.15]	-7.40 [-2.71]	-9.60 [-3.51]	-9.80 [-3.56]	-9.81 [-3.60]	-11.30 [-4.19]	-5.71 [-2.19]
rmw	-7.50 [-2.82]	0.53 [0.19]	-3.43 [-1.24]	-5.61 [-1.98]	-3.95 [-1.43]	-7.00 [-2.56]	-2.31 [-0.83]
cma	10.59 [2.61]	17.35 [4.33]	21.33 [5.32]	18.18 [4.24]	20.80 [5.18]	21.94 [5.54]	10.40 [2.54]
umd	2.96 [2.15]	3.98 [2.86]	4.56 [3.24]	4.59 [3.18]	3.97 [2.79]	3.31 [2.33]	1.18 [0.86]
# months	732	732	732	630	732	732	630
$\bar{R}^2(\%)$	19	15	12	13	12	14	26

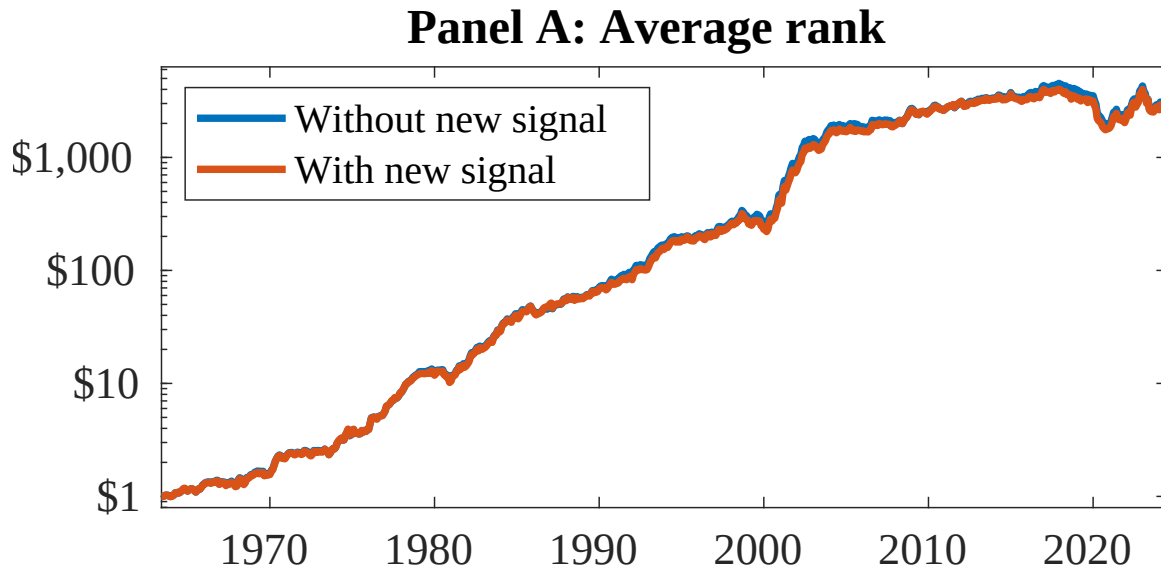


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as DCP. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Berk, J. B., Green, R. C., and Naik, V. (1999). Optimal investment, growth options, and security returns. *Journal of Finance*, 54(5):1553–1607.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. (1991). Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance*, 46(1):209–237.
- Cochrane, J. H. and Piazzesi, M. (2005). Bond risk premia. *American Economic Review*, 95(1):138–160.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1992). The cross-section of expected stock returns. *Journal of Finance*, 47(2):427–465.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gould, J., Johansen, D., and Rakshit, D. (2014). Operating leverage, profitability, and capital structure. *Journal of Financial Economics*, 114(3):424–443.
- Hennessy, C. A. and Whited, T. M. (2007). How costly is external financing? evidence from a structural estimation. *Journal of Finance*, 62(4):1705–1745.
- Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.
- Kogan, L. and Papanikolaou, D. (2013). Firm characteristics and stock returns: The role of investment-specific shocks. *Review of Financial Studies*, 26(11):2718–2759.
- Liu, L. X., Whited, T. M., and Zhang, L. (2009). Investment-based expected stock returns. *Journal of Political Economy*, 117(6):1105–1139.
- Livdan, D., Sapriz, H., and Zhang, L. (2009). Investment, capital structure, and financing constraints: A unified framework. *Journal of Finance*, 64(3):1323–1364.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.
- Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Working Paper*.
- Titman, S., Wei, K. J., and Xie, F. (2004). Capital investments and stock returns. *Journal of Financial and Quantitative Analysis*, 39(4):677–700.
- Whited, T. M. and Wu, G. (2006). Financial constraints risk. *Review of Financial Studies*, 19(2):531–559.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.